



You missed!

WWII pilots found out the hard way that if their tracers hit, their bullets missed, the ballistics were that different.



Sweaty paws?

Did you know that your kitty sweats through its paws? They're probably massaging their sweat into you when they start padding you.

Feature

LQG imagines space time to be comprised of fine fabric stitched from tiny loops, hence the name. Unlike string theory, LQG doesn't involve added dimensions.

While both the theories have their own upsides and flaws, the theory of quantum of gravity remains an unresolved question because neither of the two has been proved experimentally. Validating any of the above mentioned theories is a major challenge in experimental physics.

The theory of quantum gravity might not have any noticeable effect in our daily lives, but it will be a true testament of how far we've progressed in science and allow us to make leaps in fascinating futuristic areas such as black hole physics, time travel and wormholes.

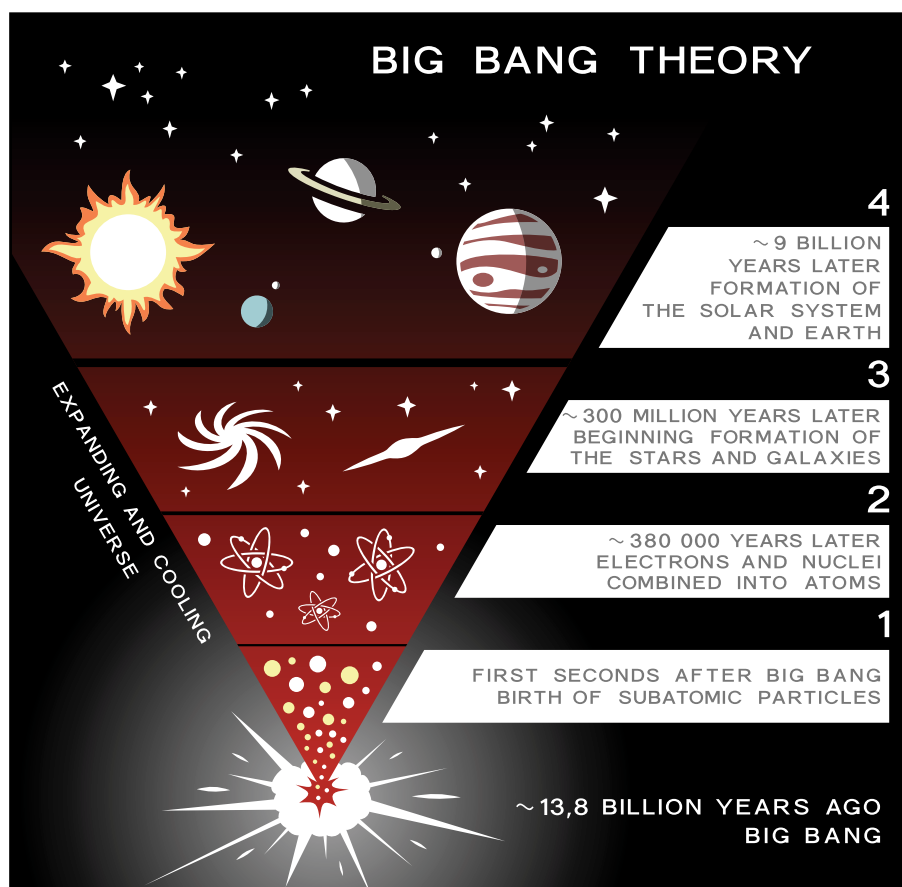
DID YOU KNOW?

Recently, scientists found that quantum gravity effects could alter the rate of electron production in the Large Hadron Collider (LHC) and this effect would occur at energies well within the range of LHC's operation. The existence of large extra dimensions would, however, have to be assumed.

Riemann Hypothesis

In an interview, distinguished number theorist, Terence Tao referred to prime numbers as the atomic elements of number theory, which of course is a spot-on characterisation. Prime numbers have no factors besides 1 and themselves and as such are the simplest construction elements of numbers. Prime numbers are also extremely erratic and follow no set pattern. A large number (product of two primes) is used to encrypt the millions of secure transactions processed online. The prime factorisation of this number currently takes practically forever. However, if we were somehow able to tame the seemingly random nature of primes and better understand how they work, we could allegedly come close to actually breaking the internet. Solving the 'Riemann Hypothesis' could bring us ten steps closer to understanding primes and could have major implications on banking, commerce and security.

As mentioned, primes aren't exactly known for being well behaved. What



A schematic of the Big bang event. A reverse big bang type event (Big Crunch) could lead to the end of the universe due to excess dark matter

Bernhard Riemann did in 1859 was find out the number of primes that are less than a given number. Riemann's solution is related to a function called the 'Riemann Zeta' function and the related distribution of points on the integer number line for which the function becomes 0. The hypothesis is concerned with a special set of such points called 'non trivial zeros', which are supposed to lie on a critical line. This hypothesis has been verified for more than a billion such zeros and could unlock the mystery surrounding the distribution of prime numbers.

Anyone in the mathematics circle will be aware that the Riemann Hypothesis is

DID YOU KNOW?

So far, only one millennium problem has been solved. In 2010, it was announced that Russian mathematician Grigori Perelman had proved the 'Poincare Conjecture'. Perelman refused to accept the million dollar prize.

one of the biggest unanswered problems there is. Not only will the solution have a huge impact on science and society, it will also guarantee the solver a million dollar prize. The Riemann hypothesis has been listed by the Clay Mathematics Institute (a private philanthropic foundation dedicated to mathematics) as one of the seven Millennium prize problems. The internet is rife with attempted proofs of the hypothesis, all of which have been rejected as valid proofs by the mathematics community. Rest assured, the internet is safe, for now.

The Survival Mechanisms of the Tardigrades

Tardigrades are a class of microorganisms found abundantly in nature in regions spanning most climatic zones and altitudes across our seven continents. But these are not your common microorganisms, they're creatures with such extreme survival skills that all those